YOLOJ: Where a Vision–Language Model Looks When It Answers Yes or No

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https://github.com/XxSamaxX/YOLOJ

Abstract

A *semantic if* reads a yes/no decision from the answer-token logits of a language model in one forward pass. We ask whether the same forward pass can also say *where*: whether a vision–language model that answers "is there a remote control?" can be turned into a detector for a predicate declared at run time, with no detection head, no label set and no training.

The obvious construction, a YOLO-like pyramid of crops with one predicate per cell, localises correctly but costs 8–16 s per image on a 27B model. Looking inside Qwen3-VL-4B instead, we find that the decision appears at layer 24 of 36, that the model attends to the image in layers 17–23, that one fixed patch absorbs 65–78% of the late attention regardless of the question, and that a handful of heads in layers 20–22 point at the object once that sink is divided out with the attention map of a content-free question. Five heads chosen on 200 COCO images hit the object on 78.5% of 200 held-out images in the pointing game (chance 10.6%, image centre 29.0%, mean of all heads 1.5%); the logit gap separates present from absent categories at AUC 0.996, and its sign is already settled at layer 24. Sharing the image prefix across questions brings each further predicate down to 12–17 ms, so all 80 COCO classes cost 1.2 s.

Turning the point into a mask is where the approach stops being competitive. Prompting SAM 2.1 or SAM 3 with the peak gives 54.5–59.5% of boxes with IoU ≥ 0.5; floating-point precision, one mask per peak and removing stray fragments do not help. SAM 3 prompted with the category *name* and no language model at all reaches 85.0%, finds 83.2% of individual instances against 35.0%, and separates present from absent at AUC 0.990. For anything that has a name, a concept segmenter is the better localiser.

What the segmenter does not do is judge. On a single photograph, its scores for "man with open mouth" and "man with closed mouth" are 0.30 and 0.54 and it scores "sleeping man" at 0.47, while the logit gap answers the same questions with margins of 16–29 nats. We therefore argue for a division of labour—the language model decides, the segmenter delineates—report a 5 Hz live demonstration and a first test on desktop screenshots, and state plainly which of these claims are measured and which rest on one image.

1 Introduction

Open implementations of the *semantic if* [10, 14] turn a language model into a conditional defined at run time: the question is posed with lettered options, a single forward pass is run, and the decision is the difference between the logits of the two letters. Ported to a vision–language model, the readout answers perceptual questions about an image in about 90 ms [15]. It says whether, not where.

Detectors of the YOLO family [17] say where, for a vocabulary fixed at training time; open-vocabulary detectors [5, 13] relax the vocabulary to noun phrases. A semantic if accepts any predicate that can be written as a question—a state, a relation, a reading of text—so a detector built on it would localise things that have no class name. This paper reports an attempt to build one, YOLOJ, on a single consumer GPU, and is as much about where the attempt loses as about where it works.

The contributions are measurements:

- A dissection of where Qwen3-VL-4B decides a yes/no question and where it looks while doing so, including an attention sink that hides the useful signal when heads are averaged (Section 5.2).
- A held-out evaluation on COCO of training-free localisation from five attention heads, and of presence and early exit from the logit gap (Section 5.3).
- A shared-prefix scheme that makes additional predicates on the same image cost 12–17 ms (Section 5.4).
- Three negative results on converting the attention peak into a mask, and a direct comparison in which SAM 3 prompted by text clearly beats the whole chain on named objects (Sections 5.5–5.6).
- Small-scale evidence on what remains for the language model: states, live video and screenshots (Section 5.7).

2 Related work

Reading decisions from logits. SemIf [10] and JEV-CPU [14] read a decision from answer-token logits; `discern` [15] carries the readout to images and studies the logit gap as a confidence signal. We reuse its prompt contract unchanged.

Attention as localisation. Zhang et al. [20] show that multimodal models attend to the relevant region even when they answer wrongly, and normalise the attention map by that of a generic instruction before cropping; our relative map is the same normalisation. Kang et al. [7] show that a few attention heads suffice for training-free grounding; we confirm this on a different model and question format and measure it against a text-prompted segmenter. FastV [3] observes that image tokens receive little attention in deep layers and prunes them. The single patch that absorbs most late attention in our measurements is reminiscent of

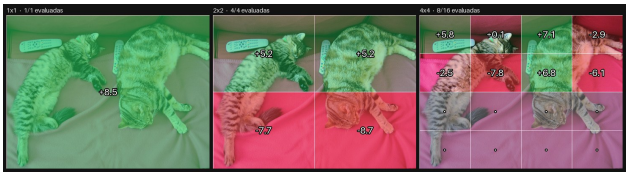


Figure 1: The pyramid of crops for "is any part of a remote control visible?": 19 forward passes, 7.7 s. Green is yes, red is no, numbers are the gap in nats; grey cells were pruned.

the artefact tokens described in vision transformers [6], although here it is observed in the language model's attention over image tokens.

Segmentation. SAM [8] and SAM 2 [16] segment from points and boxes; SAM 3 [2] adds prompts given as short noun phrases and returns every instance of the concept. LISA [9] trains a language model to emit a segmentation token for queries that need reasoning. Nothing in the present work is trained.

Benchmarks. We use COCO val2017 [12] and the pointing game [19]. Our absent categories are drawn at random, the easiest of the negative-sampling settings of POPE [11].

3 Method

3.1 Readout

The model receives a system message, the image, and a JSON object with the criterion and two options, `A`: yes and `B`: no. Generation is not run: the decision is the gap $g = \ell_A - \ell_B$ between the logits of the two letters at the first answer position, in nats. A bare object name is wrapped into a presence question; a full question is passed as written.

3.2 Pyramid of crops

The black-box construction evaluates the predicate on the whole image, then on a 2×2 and a 4×4 grid of crops, pruning the children of any cell whose gap falls below -3 nats (Figure 1). Because the model leans towards letter `A` by about 2.4 nats, a cell under 6 nats is asked again with the options swapped and the two gaps averaged. This needs only an inference server and was run against Qwen3.8-27B quantised to about 2 bits per weight.

3.3 Relative attention of a few heads

For the white-box route we run Qwen3-VL-4B [1] in `transformers`, unquantised in bfloat16, on the whole image. Its vision encoder [18] produces one token per 32×32 pixels; the language model has 36 layers of 32 heads. For a head h we take the attention from the last prompt position to the image tokens, $a_h(q)$, reshaped to the token grid. The map for a question q is

$$m(q) = \sum_{h \in H} \frac{r_h}{\sum r_h}, \qquad r_h = \frac{a_h(q)}{a_h(q_0) + \epsilon},$$

where q_0 is the content-free question "Is this an image?" asked of the same image, and H is a set of five heads. The map is smoothed with a 3×3 mean and its maximum is the predicted location. The heads are ranked on a tuning set by how well their map matches the object mask and fixed thereafter: (layer.head) 20.15, 20.4, 21.18, 21.31 and 22.6.

3.4 Shared image prefix

Every question about an image shares the start of the prompt, system message plus image. That prefix is run once and its key/value cache kept; the questions and q_0 are then a batch of suffixes of about 45 tokens each on top of the cache.

3.5 From a point to a mask

The peak of $m(q)$ is handed to SAM as a positive point. Of the three masks SAM proposes we keep the one whose footprint on the token grid has the highest IoU with the map thresholded at half its maximum. As an alternative that uses no language model, SAM 3 is prompted with the category name and its instances scoring at least 0.5 are kept.

4 Experimental setup

All runs use one RTX 3080 with 12 GB. Images come from COCO val2017. We sample 400 images with one *present* category whose instances together cover 0.5–30% of the image and one *absent* category drawn at random. The first 200 (half A) are used for every choice—heads, thresholds, number of peaks—and the other 200 (half B) only for reporting. Unless stated, a *good box* is one whose IoU with the box of the union of the real instances is at least 0.5, and results are split by whether the image has one instance of the category (134 images of half B) or several (66).

The pointing game counts a hit when the peak falls on a grid cell at least a quarter covered by the object. Chance is the mean fraction of such cells.

5 Results

5.1 The pyramid works and is too slow

On five predicates over three images the pyramid localises as expected (Figure 1), at 0.7 s per forward pass on the whole image and 0.37 s per crop. An absent object costs one forward pass; a present one costs nearly the full grid, 7.7–15.7 s, because crop gaps sit under the calibration threshold. Passing the region as text coordinates instead of a crop localised poorly. This route was read by eye only and is kept as a reference.

5.2 Where the answer is made

Reading the gap from each layer's hidden state gives noise up to layer 23, ± 7 to ± 9 nats with the right sign at layer 24, and ± 19 to ± 22 at the last layer, in all five first cases.

Map	Hit rate	AUC
Chance (object area)	10.6%	0.500
Centre of the image	29.0%	0.650
Mean of all 1152 heads, raw	1.5%	0.648
Head 20.15, raw	59.5%	0.837
Head 20.15, relative	71.5%	0.880
Five chosen heads, relative	**78.5%**	**0.892**

Table 1: Pointing game on 200 held-out COCO images. AUC is that of the map as a per-cell classifier of the object mask.

The attention to image tokens concentrates in layers 17–23, immediately before the decision. From layer 18 on, 65–78% of that attention goes to one patch that does not change with the question; averaging heads shows only that patch. A few heads in layers 20–21 nevertheless place 25–76% of their image attention inside the object, and their attention mass drops when the object is absent (Figure 2). A forward pass takes 126 ms and 8.3 GiB; returning every layer's attention adds little.

5.3 Held-out localisation on COCO

Table 1 and Figure 3 give the pointing game on half B. Averaging all 1152 heads is worse than chance because of the sink. The best single head, 20.15, was spotted by eye on two photographs in the first pass and is also the best on 198 images. Dividing by the map of the content-free question raises it from 59.5% to 71.5%; masking the three most-attended cells instead gives 59–64%. Five heads reach 78.5%.

Presence. On the same images the gap at the last layer separates the present from the absent category at AUC 0.996, and the attention mass of three chosen heads at 0.992.

Early exit. The gap read at each layer is at chance up to layer 21. Its sign is correct on 96.5% of items at layer 24 and on 96.5% at layer 35; the remaining layers sharpen the gap (AUC 0.969 to 0.996) without changing its side. We did not truncate the model to measure the saving.

5.4 The image once, the questions in a batch

The prefix of system message and image is 338 of about 385 prompt tokens. Running it once costs 133 ms (49 ms of vision encoder, 85 ms of language model); a predicate plus q_0 on top cost 34 ms, and each further predicate 12–17 ms (Table 2). The answers match those of a plain forward pass on 30 images by 3 questions: the gap moves by 0.26 nats on average and 1.0 at most, the sign agrees on 90 of 90, and the head maps correlate at 0.99 on average and 0.83 at worst. Batches above 13 questions exhaust the 12 GB when attention is returned for every layer.

5.5 From the point to a mask

Thresholding the attention map alone gives a mask IoU of 0.36 and 42% of good boxes: each cell is 32 pixels wide, so the map is a blob by construction. Table 3 gives the result of prompting a segmenter with the peak. Choosing among SAM's three candidates by agreement with the map beats choosing by SAM's own score (58.0% against 48.5% for SAM 2.1 large). The ceiling, had the best of the three candidates always been chosen, is 65.5–71.0%.

Three attempts to improve on this failed, and we report them as such.

Precision. Ragged mask borders suggested a half-precision artefact. Running the segmenters in float32 changes nothing beyond noise and costs 2.6–2.9 times the time (Table 3, lower block).

One mask per peak. Passing each peak of the map as its own object and joining the masks, with the number of peaks and their threshold chosen on half A, helps images with several instances (24% to 30%) and hurts those with one (70% to 66%): 54.5% overall against 55.0%. The eight points it gained on the tuning half did not carry over.

Removing fragments. At full resolution 39–50% of the chosen masks come in more than five connected pieces. Keeping only the piece that contains the point, or the largest, lowers good boxes by 3–6 points (Table 4). Images with one instance are unchanged; images with several lose, because the reference is the union of all instances and the "stray" pieces were often the other instances. The fragments are a cosmetic problem; the instances are the real one.

5.6 A concept segmenter does it better

SAM 3 prompted with the category name, with no language model involved, is compared with the chain in Table 5 and Figure 4. Scored against the union it gives 85.0% of good boxes against 58.0%, and the gap between one and several instances disappears (87% and 82%). Matching each real instance one-to-one to a predicted box at IoU ≥ 0.5, it finds 83.2% of 386 instances against 35.0%. On objects under 3% of the image it gives 77% of good boxes against 38%. Its maximum instance score separates

	Plain	Shared prefix
Image, predicate and q_0	259 ms	167 ms
Each further predicate	128 ms	12–17 ms
All 80 COCO classes	∼10.5 s	1.2 s

Table 2: Cost per image at 640×480, Qwen3-VL-4B in bfloat16.

Segmenter (1 point)	Mask IoU	Good boxes	One / several	ms
Attention alone	0.36	42.0%	39 / 48	–
SAM 2.1 small	0.531	54.5%	69 / 24	21
SAM 2.1 large	0.553	58.0%	74 / 26	71
SAM 3 (tracker)	0.549	59.5%	73 / 32	160
SAM 2.1 small, float32	0.537	55.5%	71 / 24	54
SAM 2.1 large, float32	0.553	58.0%	75 / 24	199
SAM 3, float32	0.551	58.0%	73 / 27	465

Table 3: Attention peak as a point prompt, held-out half, pixel level. "One / several" is the percentage of good boxes on images with one instance of the category and with several. Top block in bfloat16.

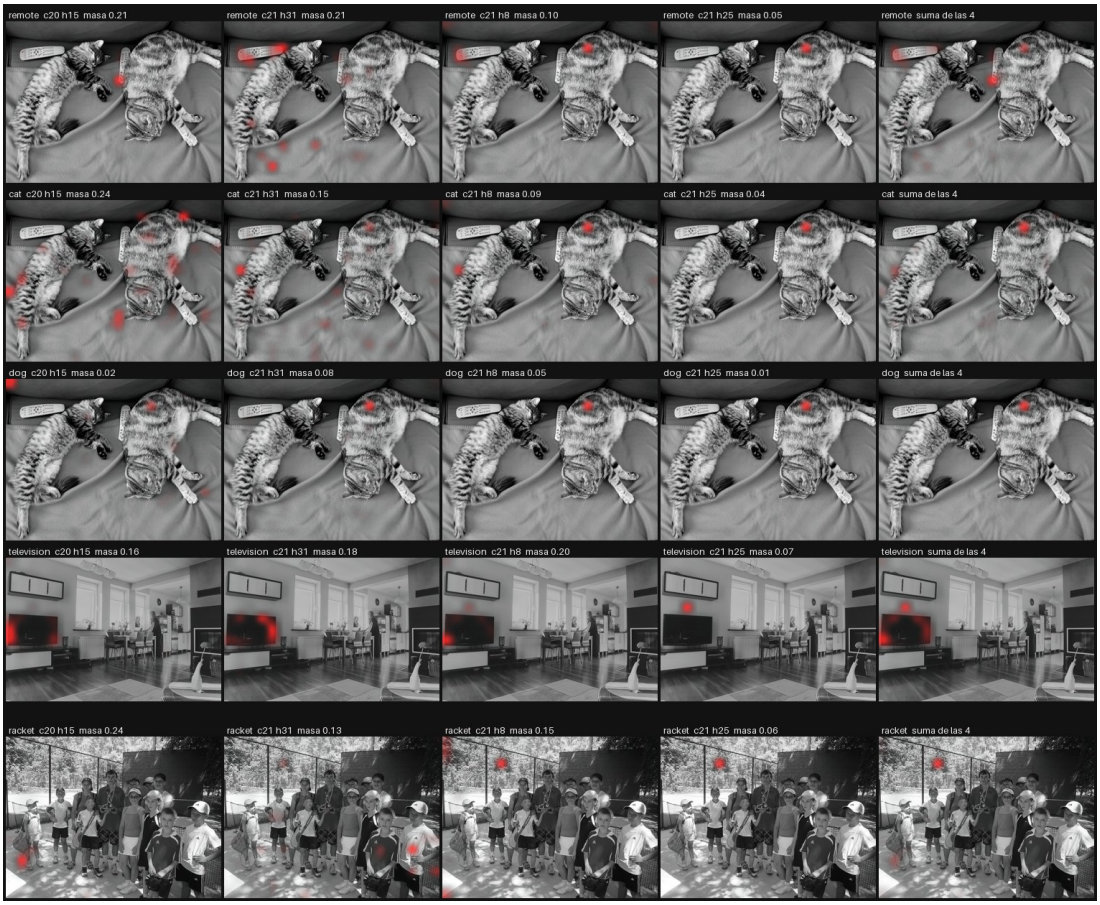


Figure 2: Raw attention of four heads (columns 1–4) and their sum (column 5) for five predicates: remote, cat, dog (absent), television, racket. Labels give layer (c) and head (h) and the attention mass on the image ("masa"). The dot that persists on the cat's back in the right-hand columns, whatever the question, is the sink. These heads were picked by eye on two of these images; the held-out evaluation is in Table 1.

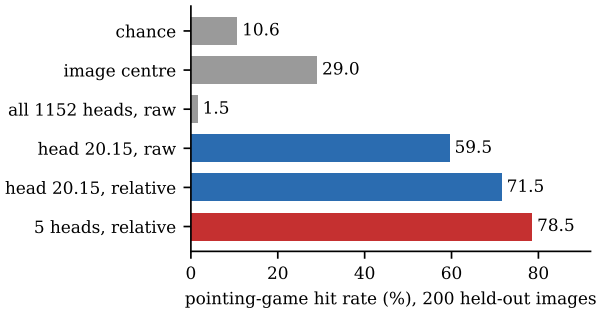


Figure 3: Pointing-game hit rate, held-out half.

the present from the absent category at AUC 0.990, against 0.996 for the logit gap. Figure 5 shows two images with several instances.

SAM 3 by text takes 187 ms per image and concept when the image is encoded each time; encoding it once takes about 160 ms and each concept after that an estimated 35–50 ms. Scanning 80 classes is therefore still several times slower than the 1.2 s of the language model, which is the one advantage the chain keeps on named objects.

Kept	2.1 small	2.1 large	SAM 3
Everything	54.5	58.0	59.5
Piece under the point	51.5	55.0	53.5
Largest piece	51.5	55.5	53.5
Pieces $\geq 10\%$ of largest	53.0	56.0	56.0

Table 4: Good boxes (%) after removing mask fragments, held-out half.

5.7 What is left for the language model

A hybrid. The results suggest a division of labour, implemented in `foto.py`: the language model decides which predicates hold; names go to SAM 3 by text; full questions, and names SAM 3 does not find, go through the attention peak. On one 1920×1080 webcam photograph, reduced to 960×540 for the language model, the 80 COCO classes take 1.9 s in total and yield person, tv and bottle with no false positive. Ten free prompts on the same photograph—six present, one question, three absent—are all answered correctly; SAM 3 delineates five of the six names and the sixth falls back to the attention peak. The three models peak at 10.8 GiB, which forces the question batch down from 9 to 4.

States. On that photograph the subject has his mouth

Method	Good boxes	Instances found	Boxes given	of them correct
Attention + point	58.0%	35.0%	200	67.5%
SAM 3 text, ≥ 0.3	81.5%	88.9%	621	55.2%
SAM 3 text, ≥ 0.5	**85.0%**	83.2%	439	73.1%
SAM 3 text, ≥ 0.7	79.5%	76.2%	345	85.2%

Table 5: SAM 3 prompted by the category name against the attention chain (SAM 2.1 large), held-out half. Crowd annotations are excluded from the instance count, which understates the last column.

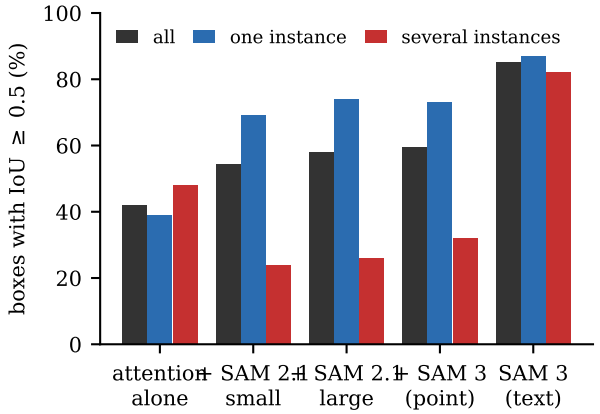


Figure 4: Boxes with IoU ≥ 0.5 by method and by number of instances of the category in the image.

closed, wears a moustache and no glasses, looks at the camera and is awake. Table 6 compares the logit gap with the score SAM 3 gives the person when the state is rephrased as a noun phrase. The segmenter returns the same person mask whatever the phrase and its score barely follows the state: it prefers "looking away" to "looking at the camera" and gives "sleeping man" 0.47 against a threshold of 0.5. The gap is right in every row with a margin of 16–29 nats, with one inconsistency, a weak yes for "looking away". This is one image, and is evidence of a direction, not a measurement.

Live video. With the shared prefix and no segmenter, a small web application marks each predicate typed by the user with a point on a webcam stream. At 640 pixels on the long side it runs about 5 analyses per second with five predicates and 6–7 with one, in 9.4 GiB. An unedited four-minute recording of a session is in the repository (`docs/demo.mp4`).

Screenshots. We marked by hand 30 interface elements and 8 absent ones on three 1920×1080 desktop screenshots, in the spirit of GUI grounding [4], and asked for each by name (Table 7, Figure 6). The model nearly always knows what is on the screen, and its point is usually near the element but seldom inside it: a 25-pixel button is smaller than one cell of the token grid. Elements of medium size are found; small toolbar buttons are not. Resolution helps on every count, so for interfaces the image should be fed as large as memory allows: 1280×720 already uses 10.4 GiB and 1600×900 does not fit.

State (truth)	SAM 3 score	Gap (nats)
Mouth open (no)	0.30	-23.2
Mouth closed (yes)	0.54	$+19.8$
Has a moustache (yes)	0.30	$+26.9$
Without a moustache (no)	0.27	–
Looking at the camera (yes)	0.07	$+16.2$
Looking away (no)	0.46	$+4.0$
Asleep (no)	0.47	-25.2
Wearing glasses (no)	0.18	-26.8
A woman (no)	0.06	-28.6

Table 6: One photograph, nine statements. SAM 3 receives a noun phrase ("man with open mouth"), the language model a question.

Long side (screen px per cell)	Present said yes	Point inside	Within 40 px	Absent said no
960 (64)	24/30	9/30	21/30	7/8
1280 (48)	26/30	10/30	23/30	8/8

Table 7: Interface elements on three screenshots. Reference boxes were drawn by hand.

6 Discussion

The signal is real and cheap. A model that was never trained to localise carries, in five of its 1152 heads, a map that hits the object four times out of five, and exposing it costs one extra short suffix per image. The sink explains why naive attention roll-ups look uninformative, and dividing by a content-free question removes it better than masking it.

It is a point, not a detector. One peak yields one object. Every attempt to recover extent and multiplicity from the same map—thresholding, several peaks, cleaning the segmenter's output—stays 25 points below a segmenter that was trained to map a noun phrase to all its instances. On named objects we see no reason to prefer the attention chain except speed when scanning many classes.

Deciding and delineating are different jobs. The segmenter's score is a poor judge of state, and the language model's attention is a poor delineator. The combination we would build next uses the gap for the condition and a segmenter for the extent, with the attention peak selecting which instance the condition is about.

Limitations. The absent categories are random and therefore easy. One category is asked per image and several instances count as one target, so none of this is a detection benchmark, and no trained detector was run for comparison. Localisation of predicates that are not object names rests on a single photograph. The screenshot references were drawn by hand by the author. The estimated per-concept cost of SAM 3 with a shared image encoding and the end-to-end time of the hybrid on COCO were not measured separately. Only one language model was dissected, and its head indices will not transfer.

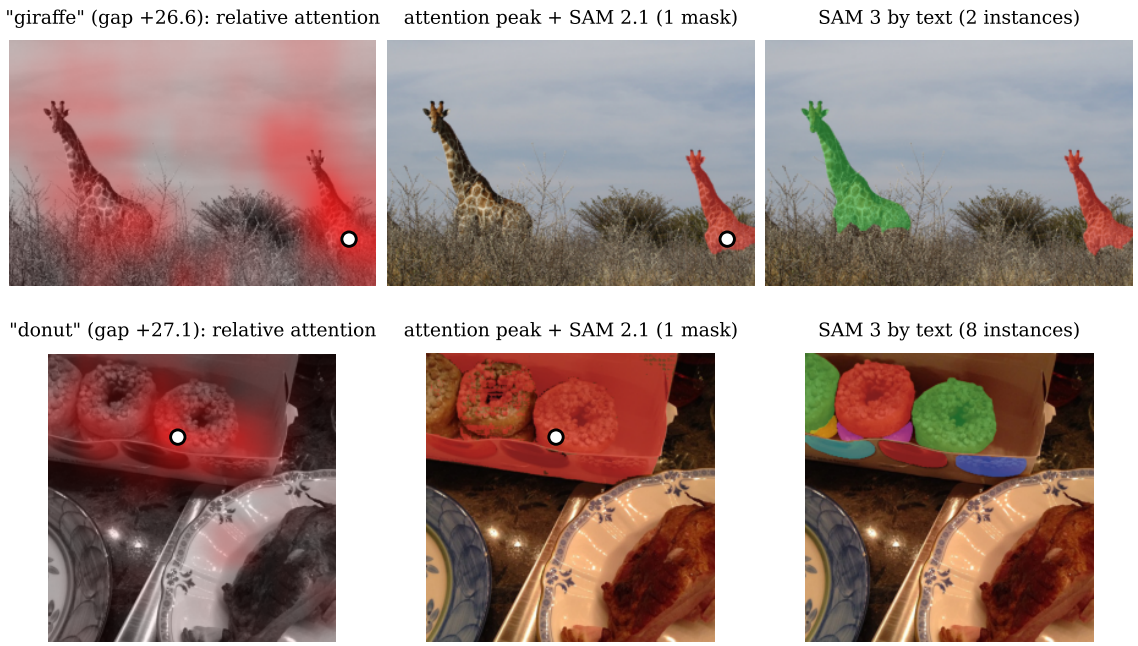


Figure 5: Two held-out images chosen to illustrate several instances. Left: the relative attention map and its peak. Centre: the mask SAM 2.1 large returns for that point. Right: SAM 3 prompted with the word alone.

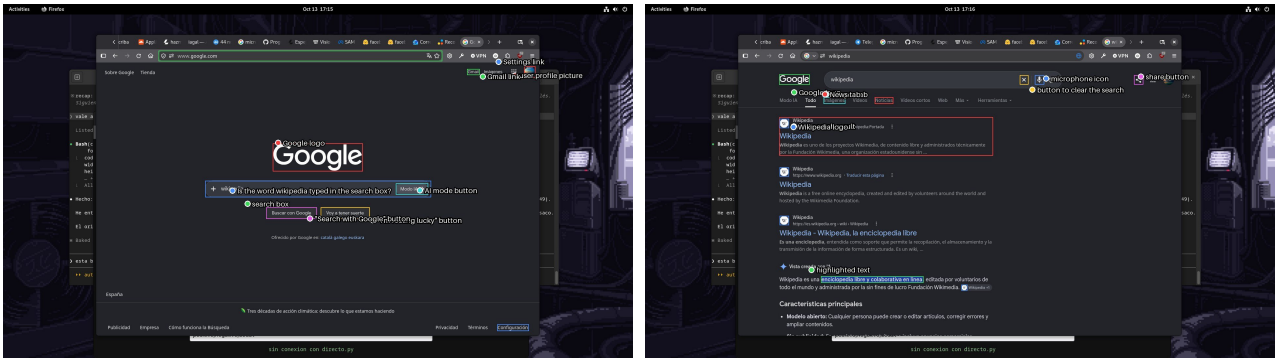


Figure 6: Two of the three screenshots at 1280 pixels. Rectangles are the hand-drawn reference; the circle of the same colour is the attention peak for that element. On these two, 16 of 18 present elements are affirmed, 5 points fall inside the element and 16 within 40 pixels.

7 Conclusions

A yes/no readout can be made to point: in Qwen3-VL-4B the decision is taken at layer 24, the model looks at the image just before it, and five heads divided by their response to a content-free question locate the object on 78.5% of held-out COCO images at no training cost and 12–17 ms per additional predicate. It cannot be made to segment competitively: SAM 3 given only the word reaches 85.0% of good boxes where the best point-prompted chain reaches 59.5%. The useful system is therefore not a language model imitating a detector but a language model judging predicates beside a segmenter that delineates them, and the experiment that would establish its worth—pairs of instances of one class that differ only in a state or relation—remains to be run.

Code and data

Code, experiment scripts, figures and this paper's sources are available at `https://github.com/XxSamaxX/YOLOJ` under the MIT licence. The prompt contract and readout come from SemIf [10] by way of `discern` [15]. COCO images keep their original licences.

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